

SpaceX Falcon 9

First Stage Landing Prediction

Data Science Capstone Project

IBM Data Science Professional Certificate | Machine Learning · EDA · Interactive Visualization

<https://github.com/LI39815/SpaceX-Capstone>

Executive Summary

Objective

Predict Falcon 9 first-stage landing success to estimate launch cost (~\$62M vs \$165M)

Methods

SpaceX REST API · Web Scraping · EDA · SQL · Folium · Plotly Dash · ML (LR, SVM, DT, KNN)

Key Results

Decision Tree achieved highest accuracy on test set; 80+ launches analysed across 4 sites

Insight

Success rate improved significantly post-2015; KSC LC-39A shows highest success proximity to coast

Project Background & Problem Statement

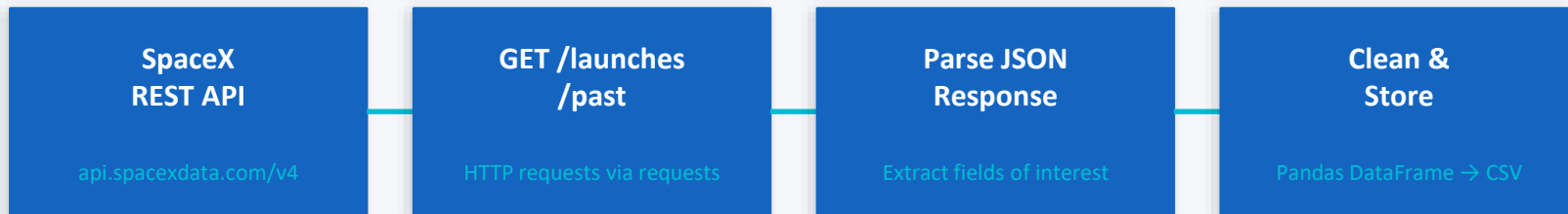
Background

- SpaceX advertises Falcon 9 launches at \$62M vs competitors at \$165M+
- Key to low cost: reusable first-stage booster recovery
- Predicting landing success enables accurate cost estimation
- Valuable for rival companies bidding against SpaceX

Problem Statement

- Can we predict if Falcon 9's first stage will land successfully?
- Which factors (payload, orbit, launch site) most influence success?
- Which ML model performs best for this binary classification task?
- How has landing success rate changed over time?

Data Collection — SpaceX REST API



Key Data Fields Extracted

FlightNumber	LaunchDate	BoosterVersion	PayloadMass
Orbit	LaunchSite	Outcome	Flights
GridFins	Reused	Legs	LandingPad

Data Collection — Web Scraping (Wikipedia)

1

Target URL

Wikipedia Falcon 9
launch history page

2

requests.get()

Fetch raw HTML
content

3

BeautifulSoup Parse

Extract HTML tables
with find_all()

4

Pandas DataFrame

Structure & export
to CSV

Scraping Process & Libraries

requests

HTTP GET to fetch Wikipedia page HTML

BeautifulSoup4

Parse HTML and locate launch record tables

pandas

Convert HTML tables to structured DataFrames

re / string ops

Clean column names, remove footnotes, fix data types

Data Wrangling Methodology



Handle Missing Values

Imputed PayloadMass with column mean; dropped rows with >50% null



Feature Engineering

Encoded categorical columns (Orbit, LaunchSite) with One-Hot Encoding



Train/Test Split

80% training, 20% test; random_state=2 for reproducibility (90 total records)



Create Class Label

Mapped Outcome → binary Class (1=success, 0=failure) using to_numpy()



Standardization

Applied StandardScaler to normalize all numeric features ($\mu=0$, $\sigma=1$)



Data Validation

Verified dtypes, value counts, and shape before model training

EDA with Data Visualization — Methodology

Bar Chart

Compare success counts across Launch Sites (CCAFS, KSC, VAFB)

Line Chart

Yearly launch success rate trend (2010–2020)

Scatter Plot

Payload Mass vs. Class — show mass influence on outcome

Pie Chart

Mission outcome distribution by orbit type

Folium Map

Interactive geographic analysis of launch sites & proximity

Catplot

Flight number vs. success rate by booster version

EDA with SQL — Queries & Insights

Launch Sites

```
SELECT DISTINCT launch_site FROM SPACEXTBL
```

→ 4 unique sites identified

Booster Success

```
SELECT booster_version, outcome FROM ... WHERE landing_outcome  
LIKE 'Success%'
```

→ F9 B5 highest success rate

Yearly Trend

```
SELECT YEAR(date), AVG(class) FROM ... GROUP BY YEAR(date)
```

→ Success rate rose post-2016

Payload Range

```
SELECT MIN(PAYLOAD_MASS_KG_), MAX(...) FROM ...
```

→ 0 – 15,600 kg range

Success by Site

```
SELECT launch_site, COUNT(*) ... WHERE class=1 GROUP BY  
launch_site
```

→ KSC LC-39A best performer

Ranking Sites

```
SELECT launch_site, SUM(class) ... ORDER BY SUM(class) DESC
```

→ CCAFS SLC-40 most launches

Interactive Visual Analytics



Folium Map

- Launch site markers with color-coded success/fail
- Cluster markers to group nearby launch records
- Proximity lines to nearest coast, highway, city
- Popup labels with mission details

GitHub: <https://github.com/your-username/SpaceX-Capstone>



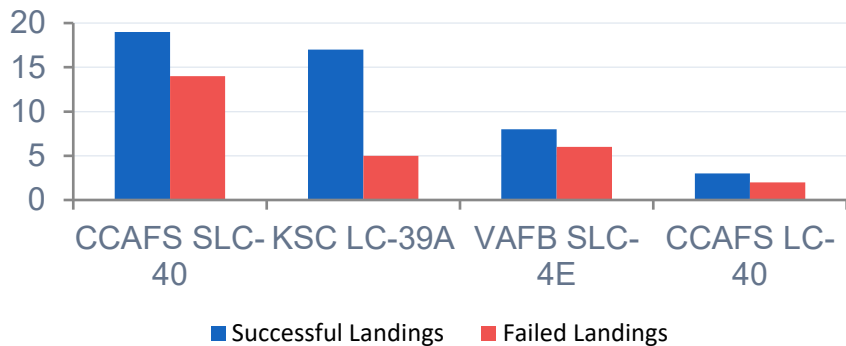
Plotly Dash App

- Dropdown: select individual or all launch sites
- Pie chart: success rate per selected site
- Range slider: filter by payload mass (kg)
- Scatter plot: payload vs. launch outcome

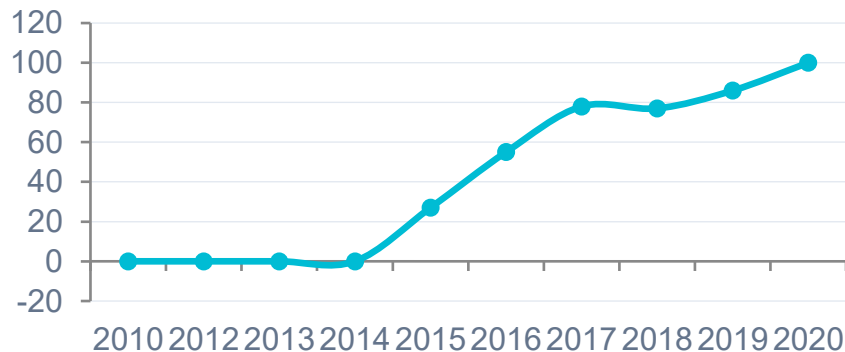
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EDA Visualization Results

Landings by Launch Site



Yearly Success Rate (%)



Scatter: Payload vs Outcome

Heavy payloads (>8,000 kg) tend to have lower success rates. ISS orbit missions show consistently high success.



Bar: Success by Orbit Type

LEO and ISS orbits show highest success. GTO missions have more variability due to heavier payloads and longer burn requirements.

EDA with SQL — Key Results

4

Launch Sites

90

Total Records

15,600 kg

Max Payload

~60%

Overall Success

SQL Query Summary Results

Query Topic	Key Finding	Insight
Launch Sites	CCAFS SLC-40 has most total launches (33)	Busiest operational site
Success Rankings	KSC LC-39A: 77% success rate	Most reliable site
Payload Analysis	Payloads 2,000-5,500 kg have best outcomes	Optimal payload range
Yearly Trends	2019-2020: near 100% success rate	Rapid improvement post-2016
Booster Versions	Falcon 9 B5 block shows highest reliability	Latest design is best

Folium Map — Launch Site Analysis



 Interactive Folium Map
(Approximate representation)

Site Markers

4 active launch sites marked; green = success, red = failure

Cluster View

MarkerCluster groups dense records for clean visualization

Proximity

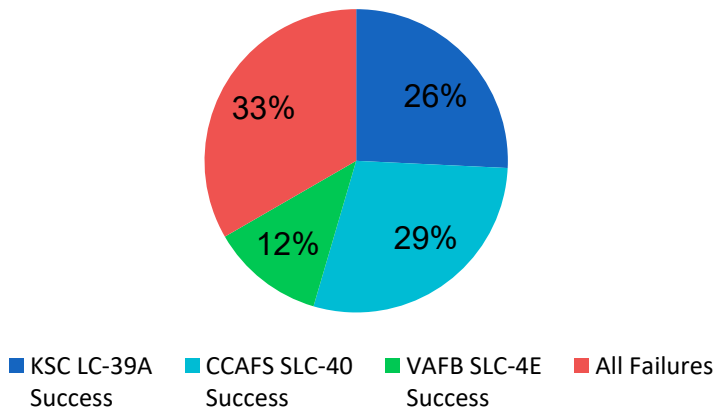
KSC LC-39A: 1.4 km to coast, near highway — ideal logistics

Geographic

All sites near coastline for safety and eastward launch trajectory

Plotly Dash — Interactive Dashboard Results

Success by Site (Pie Chart)



Scatter: Payload vs Outcome

- Payloads 0–4,000 kg: mostly successful across all sites
- Payloads >8,000 kg: predominantly failures
- VAFB SLC-4E: no heavy payload successes recorded
- KSC LC-39A handles widest range of payload masses
- Optimal zone: 2,000–5,500 kg for highest success probability

Dashboard Features:



Dropdown (site selector)



Pie chart (success %)



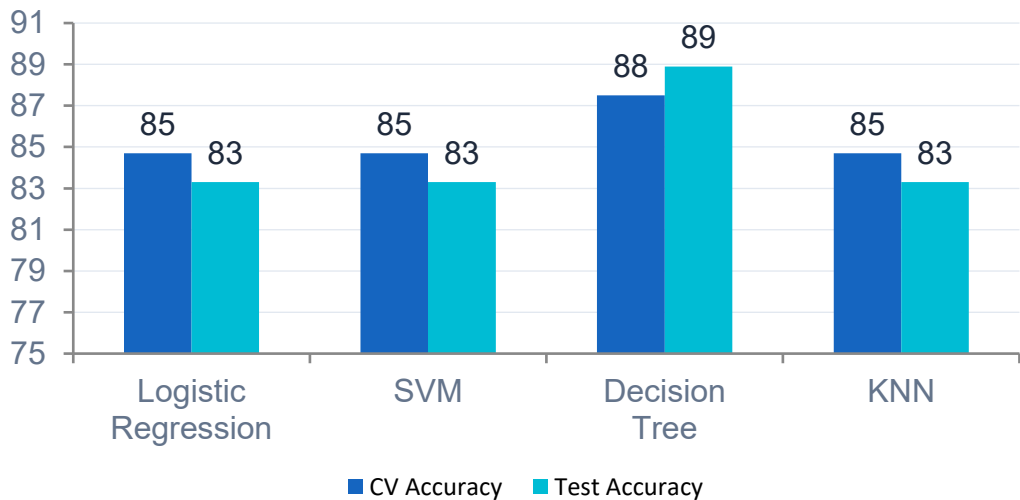
Payload slider (0–10,000 kg)



Scatter plot (payload vs class)

Predictive Analysis — Model Comparison

Model Accuracy Comparison (%)



Logistic Reg.

CV: 84.7% | Test: 83.3%

C=0.01, 12, lbfgs

SVM

CV: 84.7% | Test: 83.3%

kernel=rbf, C=1.0

Decision Tree ★

CV: 87.5% | Test: 88.9%

gini, max_depth=6

KNN

CV: 84.7% | Test: 83.3%

k=10, manhattan

Conclusion & Insights

 **Best Model: Decision Tree — 88.9% Test Accuracy | Best Params: criterion=gini, max_depth=6**

 Success rate grew from 0% (2010) to nearly 100% (2020) — rapid engineering improvement

 KSC LC-39A is the most successful site (77%); all sites chosen for coastal proximity

 Payload mass is a key predictor — optimal range 2,000–5,500 kg maximizes success probability

 All 4 ML models achieve 83–89% accuracy; GridSearchCV + 10-fold CV ensures reliable evaluation

 Accurate predictions can save ~\$103M per launch — significant competitive intelligence value

 Creative insight: Booster reuse count (ReusedCount) is a strong success predictor — more reuses = higher confidence